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Hotel Management System

Experiment - 3

SRS: Software Requirements Specifications

1. INTRODUCTION

This system is designed in favour of the hotel management, which helps them to save the records of the students about their rooms and other things. It helps them from the manual work from which it is very difficult to find the record of the students and the mess bills of the students and the information about those who had left the hostel three years before. This system automatically calculated all the bills and issued notifications for those students who were against some rules.

1.1. Purpose

The Software Requirements Specification (SRS) will provide a detailed description of the Hotel Management System (HMS) requirements. This SRS will allow for a complete understanding of what is to be expected from the newly introduced system which is to be constructed. A clear understanding of the system and its' functionality will allow for the correct software to be developed for the end user and will be used for the development of the future stages of the project. This SRS will provide the foundation for the project. From this SRS, the Hotel Management System can be designed, constructed, and finally tested.

This SRS will be used by the system development team which is constructing the HMS and the hotel end users. The Project team will use the SRS to fully understand the expectations of this HMS to construct the appropriate software. The hotel end users will be able to use this SRS as a "test" to see if the constructing team will be constructing the system to their expectations. If it is not to their expectations, the end users can specify how it is not to their liking, and the team will change the SRS to fit the end users' needs.

1.2. Intended Audience and Reading Suggestions

The intended audience of this document would be the owner and specific employees like the Manager and Receptionist of Hotel XYZ and the project team to refer and analyze the information. The SRS document can be used in any case regarding the requirements of the project and the solutions that have been taken. The document would finally provide a clear idea about the system that is building.

A brief outline of the document is:

1. Overall Description

2. System Features

3. External Interface Requirements

4. Non-Functional Requirements

1.3. Scope

The introducing software, Hotel Management System, which will be implemented for Hotel XYZ, will automate the major operations of the hotel. The Reservation System is to keep track in room and hall reservations and check availability. The Room Management System is for managing all room types and room services. The Inventory Control System will keep track in all inventories of the hotel, and guest details will be handled by guest management. The administration department will monitor all. There are three End Users for HMS. The End Users are the Owner, Manager and Receptionist. The owner can access all system functionalities without any restrictions. Managers can access all system functionalities with limited restrictions. The receptionist can only access the Reservation management section. HMS can create different Login functions to keep restrictions for each End User level.

The objective of the automated Hotel Management System is to simplify the day-to-day processes of the hotel. The system will be able to handle many services to take care of all customers in a quick manner. As a solution to a large amount of file handling happening at the hotel, this software will be used to overcome those drawbacks. Safety, easiness of use and most importantly the efficiency of information retrieval are some benefits the development team is going to present with this system. The system should be user appropriate, easy to use, provide easy recovery of errors and have an overall end-user high subjective satisfaction.

1.4. Definitions & Abbreviations

- SRS: Software requirement specification
- HMS: Hotel management system
- System operator: System administrator, library staff, data entry operator
- RAM: Random access memory
- MERN: MongoDB, Express.js, React.js, Node.js tech stack.
- Hostel staff: The user has the privilege to issue, return, reserve and query books.

1.5. References

Books:

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[2] ElmasriNavathe,*Fundamentals of Database System 3rd edition*.Pearson education,2000.

[3] RaguRamakrishnan/JohnesGehrke,*Database Management Systems 3rd edition*.McGrow-HILL,2003.

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[1] ”Hotel Management Case Study”, March.6, 2010. [Online].Available:

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[3] Fernandez & Yuan, X,(1999). *An analysis Pattern for Reservation and Use of Reusable Entities*. PloP 1999 conference, Retrieved from **http://hillside.net/plop/plop99/proceedings/Fernandez2/reservanalysisPattern3.PDf**

2. OVERALL DESCRIPTION

2.1. Product Perspective

The Hotel Management System is a new self-contained software product which will be produced by the project team to overcome the problems that have occurred due to the current manual system. The newly introduced system will provide easy access to the system, and it will contain user-friendly functions with attractive interfaces. The system will give better options for the problem of handling large-scale the physical file systems, for the errors occurring in calculations and all the other required tasks that have been specified by the client. The final outcome of this project will increase the efficiency of almost all the tasks done at the Hotel in a much more convenient manner.

2.1.1. Hardware Interfaces

The requirements of a desktop computer are to access the web browser where the system will run. A specific computer must match with the above-mentioned requirements to gain the maximum benefits from the system efficiently. Reservation alerts will be sent to one of the member of the hotel staff as e-mail notification. So there is a need of a broadband internet connection. Client should able to keep a stable internet connection. A printer will be needed when printing bills and several reports.

- Intel x86 or compatible processor
- Minimum of 512 MB RAM
- Minimum of 50 MB hard drive space
- TCP/IP protocol support
- Compatible operating systems:
- An x86 Linux operating system.
- A 64-bit Windows operating system such as Windows Vista, Windows 7, Windows 8, Windows 10, Windows Server 2008 or Windows Server 2012.
- An OS X operating system x64.

2.1.2. Software Interfaces

The MERN stack is a complete open source combination, all JavaScript-related technologies are also the hottest today: MongoDB, Express.js, React / React Native, NodeJS.

MongoDB is an open source database; it is also the leading NoSQL (*) database currently used by millions of people. It is written in one of the most popular programming languages today. In addition, MongoDB is cross-platform data that operates on the concepts of Collections and Documents, providing high performance with high availability and ease of expansion.

Express.js is a framework built on top of Nodejs. It provides powerful features for web or mobile development. Express.js supports HTTP and middleware methods, making the API extremely powerful and easy to use.

React. js is an open-source JavaScript library used to build user interfaces specifically for single-page applications. It's used for handling the view layer for web and mobile apps.

Node.js is an open source, a system application and furthermore, is an environment for servers. Nodejs is an independent development platform built on Chrome's JavaScript Runtime that we can build network applications quickly and easily. Google V8 JavaScript engine is used by Node.js to execute code. Moreover, a huge proportion of essential modules are written in JavaScript.

2.1.3. Communications Interfaces

When a specific reservation reserved at the same time an e-mail notification will be sent to both relevant staff member's e-mail account and guest's account. Guest will be notified in the check-out date. To achieve that functionality, it requires having a stable internet connection. Mostly a broadband connection with the client's computer will provide the efficient service.

2.1.4. User Interfaces

- User-friendly dashboard of system.
- A login interface is used to log in to the system using username and password for three different users.
- Adding new guest to the system.
- Make a new reservation.
- View reservations.

2.2. Product Functions

- Make Reservations
- Search Rooms
- Add Payment
- Issue Bills
- Manage Guest (Add, Update Guest)
- Manage Room Details (Add, Update, Delete)
- Manage Staff (Add, Update, Delete, View)

- Manage Inventory (Add, Edit, Delete)
- Set Rates
- Retrieve Reports (Staff payment, Income)
- Manage Users (Add, Update, Delete)

2.3. User Characteristics

Admin:

Hotel owner has the privilege of Monitoring and authorization of all the tasks handle by the system. He can access every function performed by the system. Owner of the company as well as the system can access to the administration panel which is consider the core of the system. As the main authorized person of the company owner gets the ability to manage the other users including their user levels and privileges. Taking backups of the system and restoring system can also be done by the Owner. Meanwhile he will be able to take all the kinds of reports available in the system. As the owner of the system and the company he has the power to set room rates as well. Hotel owner has the sole right of deleting a staff member from the system database.

Manager:

Manager is responsible for managing resources available in hotel management system. Manager also has most of the privileges mentioned above except the things regarding the payment handling. The reason for using a Manager is to reduce the work load done by the owner that cannot be assigned to the receptionist, as those tasks seem much responsible. The user level, Manager has the authority to take all the reports available in the system but here also except the reports related to financial stuff, hotel income. Manager has other abilities that receptionist, user level has. Such as, adding new staff member to the system, Modifying them or removing them, Adding new guests to the system, Modifying them and removing them from the system, Adding new inventory to the system, Modifying them and removing them. Adding new room types to the system, modifying them and removing them

Receptionist:

As a hotel receptionist, he or her role will be to attain the goals of bookings and to ensure that all guests are treated with a high standard of customer service. Hierarchically receptionist role has the least accessibility to the system functions. Receptionist plays the boundary role of the system .He or she can perform limited functions such as registering new guest to the system, make reservations, Sending e-mail reminders to clients for booking confirmation.

2.4. Constraints

Software development crew provides their best effort in developing the system. In order to maintain the reliability and durability of system, some design and implementation constraints are applied. Availability of an android app for hotel management system could make the system portable but due to time constraint it is not possible. System will need a minimum memory of 512MB. But it is recommended to have a memory of 1GB. When designing interfaces of system, we had the capability of work with new tools such as Dev Express. Considering the client's budget we decided to create those interfaces in a simple realistic manner using affordable technology.

2.5. Assumptions and Dependencies

Some software used in implementing the system is with high cost and the client has agreed to afford the amount of money needed to purchase them. It's assumed that the client won't change that decision on the next phases of the software development. Although we assume that the client is using windows 7,8 or 10. Otherwise, if the client uses an open-source operating system, there is a need to change the SRS accordingly.

3. SPECIFIC REQUIREMENTS

This system is designed in favour of the hostel management, which helps them to save the records of the students about their rooms and other things. It helps them from the manual work from which it is very difficult to find the record of the students and the mess bills of the students and the information about those who had left the hostel three years before. This system automatically calculated all the bills and issued notifications for those students who were against some rules.

3.1. External Interfaces

The Hotel Management System will use the standard input/output devices for a personal computer. This includes the following:

- Keyboard
- Mouse
- Monitor
- Printer

3.1.1. User Interfaces

- User friendly dashboard of system.
- Login interface is used to login to the system using username and password for three different users.
- Adding new guest to the system.
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- View reservations.

3.1.2. Software Interfaces

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3.2. Functions

- Make Reservations
- Search Rooms
- Add Payment
- Issue Bills
- Manage Guest (Add, Update Guest)
- Manage Room Details (Add, Update, Delete)
- Manage Staff (Add, Update, Delete, View)
- Manage Inventory (Add, Edit, Delete)
- Set Rates
- Retrieve Reports (Staff payment, Income)
- Manage Users (Add, Update, Delete)

3.3. Performance Requirements

Performance requirements define acceptable response times for system functionality.

- The load time for user interface screens shall take no longer than two seconds.
- The log in information shall be verified within five seconds.
- Queries shall return results within five seconds.

3.4. Logical Database Requirements

The logical database requirements include the retention of the following data elements. This list is not a complete list and is designed as a starting point for development.

<u>Booking/Reservation System</u>

- Customer-first name

- Customer last name
- Customer address
- Customer phone number
- Number of occupants
- Assigned room
- Default room rate
- Rate description
- Guaranteed room (yes/no)
- Credit card number
- Confirmation number
- Automatic cancellation date
- Check-in date
- Check-out date
- Customer feedback
- Payment received (yes/no)
- Payment type
- Total Bill

<u>Food Services</u>

- Meal
- Meal type
- Meal item
- Meal order
- Meal payment (Bill to room/Credit/Check/Cash)

3.5. Design Constraints

3.5.1. Standards Compliance

There shall be consistency in variable names within the system. The graphical user interface shall have a consistent look and feel.

3.6. Software System Attributes

3.6.1. Reliability

Specify the factors required to establish the required reliability of the software system at time of delivery.

3.6.2. Availability

The system shall be available at all hours of operation.

3.6.3. Security

Customer Service Representatives and Managers will be able to log in to the Hotel Management System. Customer Service Representatives will have access to the Reservation/Booking and Food subsystems. Managers will have access to the Management subsystem as well as the Reservation/Booking and Food subsystems. Access to the various subsystems will be protected by a user log in screen that requires a user name and password.

3.6.4. Maintainability

The Hotel Management System is being developed in MERN Stack.

3.6.5. Portability

The Hotel Management System shall run in any desktop environment which supports a browser that can run MERN Stack based web applications.

3.7. Organizing the specific requirements

3.7.1. System mode

The system performs as expected in all supported browsers in online mode.

3.7.2. User Class

Admin: Admin has the privilege of Monitoring and authorization of all the tasks handle by the system. He can access every function performed by the system. Owner of the company as well as the system can access to the administration panel which is consider the core of the system. As the main authorized person of the company owner gets the ability to manage the other users including their user levels and privileges. Taking backups of the system and restoring system can also be done by the Owner. Meanwhile he will be able to take all the kinds of reports available in the system. As the owner of the system and the company he has the power to set room rates as well. Hotel owner has the sole right of deleting a staff member from the system database.

Manager: Manager is responsible for managing resources available in hotel management system.

Manager also has most of the privileges mentioned above except the things regarding the payment handling. The reason for using a Manager is to reduce the work load done by the owner that cannot be assigned to the receptionist, as those tasks seem much responsible. The user level, Manager has the authority to take all the reports available in the system but here also except the reports related to financial stuff, hotel income. Manager has other abilities that the receptionist, user level has. Such as, adding new staff member to the system, Modifying them or removing them, Adding new guests to the system, Modifying them and removing them from the system, Adding new inventory to the system, Modifying them and removing them. Adding new room types to the system, modifying them and removing them.

Receptionist: As a hotel receptionist, he or her role will be to attain the goals of bookings and to ensure that all guests are treated with a high standard of customer service. Hierarchically receptionist role has the least accessibility to the system functions. Receptionist plays the boundary role of the system. He or she can perform limited functions such as registering new guest to the system, make reservations, Sending e-mail reminders to clients for booking confirmation.

3.7.3. Feature

a) Validity Checks: JavaScript provides validity checks for various fields in the forms.

b) Sequencing Information: All the information regarding room details, guest list, food details, and display of booking details should be handled sequentially that is, data should be stored only in a particular sequence to avoid any inconvenience

c) Error Handling: If any validations or sequencing flows do not hold true, then appropriate error messages will be prompted to the user for doing the needful.

3.7.4. Functional Hierarchy

Hierarchically at the top, admin has all the authority to manage the other users including their user levels and privileges. Taking backups of the system and restoring system can also be done by the owner. Meanwhile he will be able to take all the kinds of reports available in the system. As the owner of the system and the company he has the power to set room rates as well. Hotel owner has the sole right of deleting a staff member from the system database.

Hierarchically at mid, manager has the authority to take all the reports available in the system but here also except the reports related to financial stuff, hotel income. Manager has other abilities that the receptionist, user level has. Such as, adding new staff member to the system, Modifying them or removing them, Adding new guests to the system, Modifying them and removing them from the system, Adding new inventory to the system, Modifying them and removing them. Adding new room types to the system, modifying them and removing them.

Hierarchically at the lowest, receptionist role has the least accessibility to the system functions. Receptionist plays the boundary role of the system. He or she can perform limited functions such as registering new guest to the system, make reservations, Sending e-mail reminders to clients for booking confirmation.

3.8. Additional Comments

We may choose any one organization and the selection is dependent on the developer's expertise, previous experience, customer's expectations, available technologies, etc. Sometimes, more than one organization may also be used for organizing section 3 of the SRS document.

4. SUPPORTING INFORMATION

A system context diagram as well as use cases and use case descriptions, table of contents have been developed in separate documents.

4.1. Table of Contents